

TEAM 7 TECHNICAL BRIEF

1. Tooling Rationale

The Farmer RAG Advisory System, Omulimi integrates three external tools to provide actionable, timely, and evidence-based agricultural guidance for East African smallholder farmers. Each tool addresses a critical information gap that cannot be satisfied through retrieval alone.

1.1 Weather Tool (Open-Meteo API)

Necessity: Agricultural decisions are fundamentally weather-dependent. While RAG provides general best practices such as maize requires 500-800mm rainfall, it cannot provide real-time precipitation forecasts, temperature ranges, or water balance calculations needed for irrigation scheduling, pest management, and harvest timing. This tool bridges the gap between static agronomic knowledge and dynamic environmental conditions.

1.2 Market Prices Tool

Necessity: Market prices fluctuate based on seasonal supply/demand, regional conditions, and global trends. Static documents cannot advise on optimal selling times or which crops offer better margins in the current market. This tool enables revenue-optimizing recommendations that combine agronomic feasibility from RAG with economic viability from market data.

1.3 Knowledge Base Tool (RAG)

Necessity: While the base LLM has general agricultural knowledge, it lacks localized best practices for East African soil types, rainfall patterns, and farming systems. This tool grounds responses in authoritative extension documents such as Ministry of Agriculture guidelines, CGIAR research, NGO protocols with source citations that build farmer trust and enable verification.

1.4 Tool Integration

Example Query: "Should I plant maize this week?"

- **RAG:** "Maize requires 50-100mm rainfall in first 3 weeks" (Rwanda MoA)
- **Weather:** 7-day forecast shows 60mm expected rainfall
- **Market:** Maize prices trending upward (+15%)

Response: "Yes, conditions favorable. Adequate rainfall forecast for germination, rising prices make it economically viable. Use SC627 variety (MoA Guide, p.14)."

2. Failure Analysis

2.1 The Failure

Symptom: RAG retrieval consistently returned zero results despite 8 ingested PDFs that amount to ~450 chunks in ChromaDB.

Test: "What fertilizer for maize at planting?" returning an [] (empty) set.

Impact: Agent could not ground responses in sources, fell back to generic LLM knowledge, failed verification (unmeasurable groundedness score).

2.2 Root Cause

Through systematic debugging, we identified three interrelated issues:

- First, Overly Strict Metadata Filters: Chunks tagged crop type: "legumes" didn't match queries for "beans" which is a perfect semantic match but filtered out by metadata mismatch.
- Second, High Similarity Threshold: threshold=0.85 excluded relevant chunks for example "fertilizer timing" vs. "nutrient application" scored 0.78 and it was rejected.

- **Third, Unsupported Filter Operators:** Used *\$regex* patterns not supported by ChromaDB and this was one of our few silent failures.

2.3 Technical Fix

Implemented a three-tier fallback retrieval strategy:

- **Tier 1 (Precision):** Filtered search with metadata (crop, stage) using equality operators only, threshold ≥ 0.75 . If zero results, then we move onto Tier 2.
- **Tier 2 (Recall):** Remove all filters, rely on semantic similarity, same threshold. If all below threshold then move onto Tier 3.
- **Tier 3 (Coverage):** Return top-k results regardless of similarity score to ensure the agent always has context to work with. These results are flagged as low confidence in metadata to signal they may be less relevant, enabling downstream human-in-the-loop review before presenting recommendations to farmers.

Additional fixes: (1) Simplified filter syntax to ChromaDB-supported operators

(2) Standardized metadata tags during ingestion

(3) Added structured logging to track tier triggers.

2.4 Results

- **Retrieval Success:** From ~0% to >85% on test queries
- **Groundedness Score:** 0.78 average which was previously unmeasurable
- **Tier Distribution:** 60% Tier 1, 25% Tier 2, 15% Tier 3
- **Quality:** Responses now include source citations, verifiably grounded in authoritative materials

3. Individual Contribution

Glorry led the core agent design and system orchestration. This included finalizing the ReAct-based agent graph and tool-calling logic, configuring and validating LangSmith tracing, implementing retrieval fallback and threshold logic, and integrating the full pipeline for end-to-end testing. Glorry also authored the System Architecture.

Ethel was responsible for the retrieval, ingestion, and verification components of the system. This included improving and validating semantic chunking during document ingestion, testing ingestion and retrieval quality across multiple documents and queries and documenting the retrieval design decisions. Ethel also reviewed and verified tool schemas and outputs, strengthened the groundedness and verification logic, and authored the Retrieval Module and Verification Module sections and prepared the technical brief.